Exploratory Data Analysis Report

Course Name & Code : Fundamentals of Data Science (DSA04)

EDA on 5 datasets: Education,Healthcare,Retail, Banking,and Agriculture.

Question 1: Education Analytics Dataset

Goal: Analyze student performance based on attendance, study hours, and internal marks to find out what affects Result (Pass/Fail).

Column Name	Data Type
Student_ID	str
Attendance	int64
Study_Hours	int64
Internal_Marks	int64
Result	str

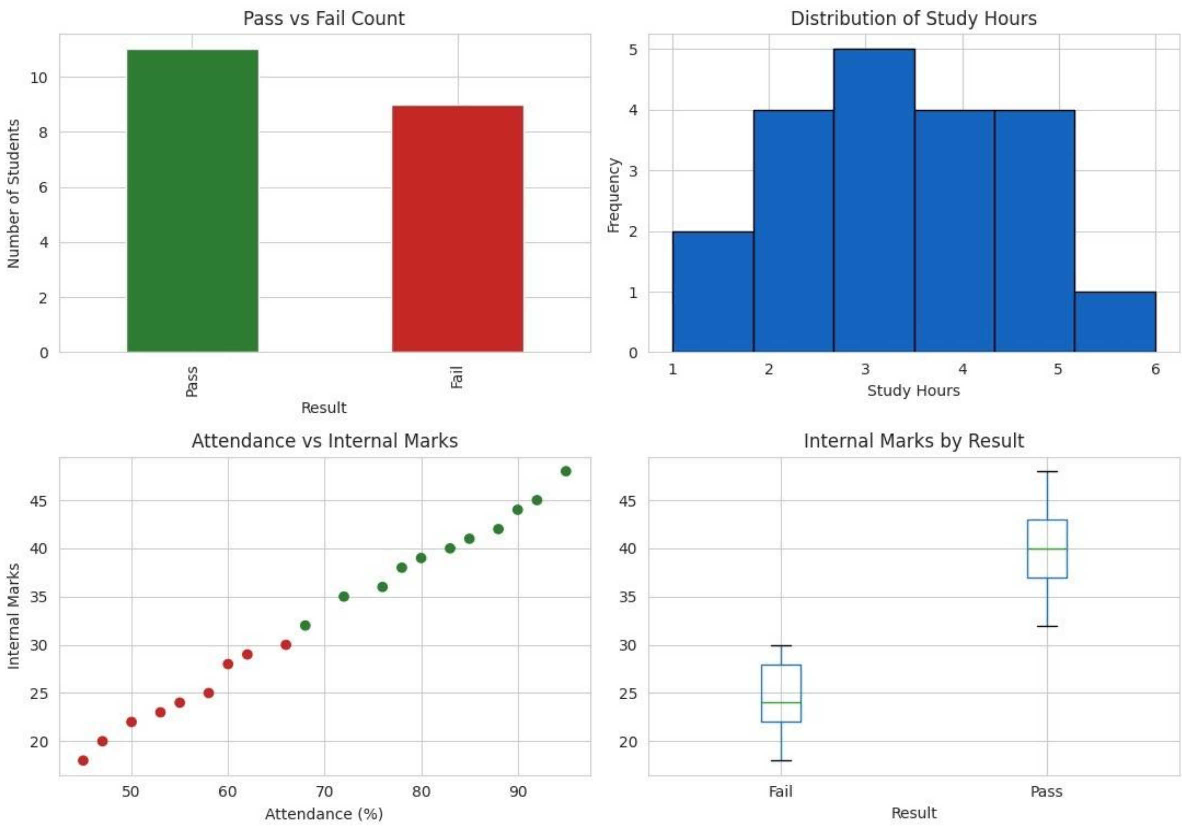
Dataset Information

Property	Value
Shape (Rows, Columns)	(20, 5)
Total Rows	20
Total Columns	5
Missing Values	0
Duplicate Rows	0

Statistic	Attendance	Study_Hours	Internal_Marks
Count	20.0	20.0	20.0
Mean	70.15	3.35	32.95
Standard Deviation (Std)	15.94	1.42	9.09
Minimum (Min)	45.0	1.0	18.0

Descriptive Statistics

Statistic	Attendance	Study_Hours	Internal_Marks
25% (Q1)	57.25	2.00	24.75
50% (Median)	70.00	3.00	33.50
75% (Q3)	83.50	4.25	40.25
Maximum (Max)	95.00	6.00	48.00



Observations

Observation No.	Description
1	Students with attendance above 75% mostly pass.
2	Higher study hours generally lead to higher internal marks.
3	Students who fail usually have internal marks below 30.
4	Attendance and internal marks show a positive relationship.
5	No missing values or duplicate records were found in the dataset.

Conclusion

Attendance and study hours are strong indicators of student performance. Students with higher attendance and more consistent study hours generally achieve better internal marks and are more likely to pass. The dataset also contains no missing values or duplicate records, indicating good data quality. Therefore, students should be encouraged to maintain high attendance and regular study habits to improve their academic performance.

Question 2: Healthcare Analytics Dataset

Goal: Analyze patient health risk based on age, sugar level, BP, and BMI to identify patterns related to disease status.

Dataset Information

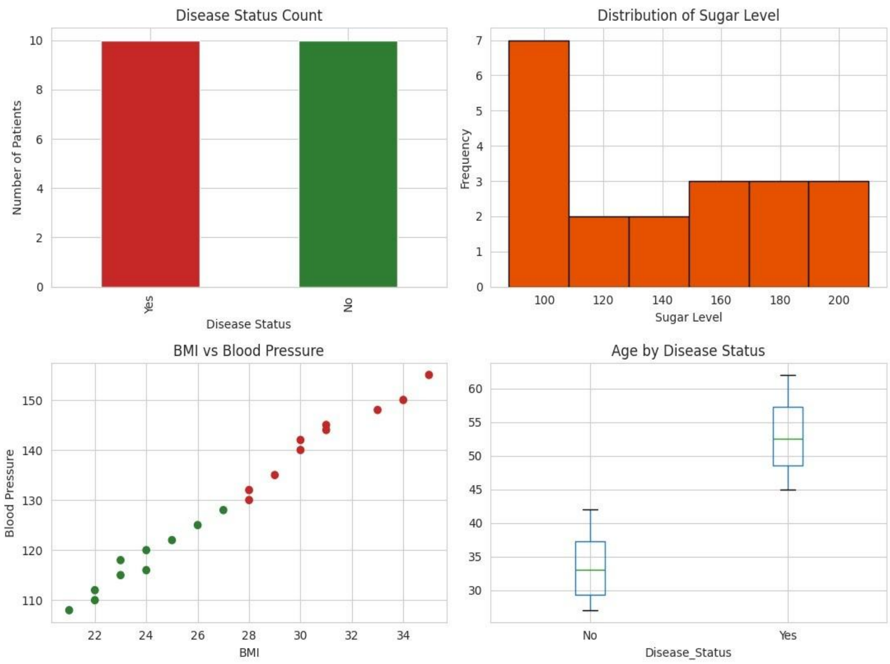
Property	Value
Shape (Rows, Columns)	(20, 6)
Total Rows	20
Total Columns	6
Missing Values	0
Duplicate Rows	0

Column Data Types

Column Name	Data Type
Patient_ID	str
Age	int64
Sugar_Level	int64
BP	int64
BMI	int64
Disease_Status	str

Descriptive Statistics (Numeric Columns)

Statistic	Age	Sugar_Level	BP	BMI
Count	20.0	20.0	20.0	20.0
Mean	43.20	138.65	129.75	27.30
Standard Deviation (Std)	11.32	40.33	14.52	4.26
Minimum (Min)	27.00	88.00	108.00	21.00
25% (Q1)	33.50	99.50	117.50	23.75
50% (Median)	43.50	137.50	129.00	27.50
75% (Q3)	52.25	171.25	142.50	30.25
Maximum (Max)	62.00	210.00	155.00	35.00



Observations

Observation No.	Description
1	Patients with sugar levels above 140 mostly have **Disease_Status = Yes**.
2	Higher BMI is associated with higher blood pressure.
3	Older patients (above 45 years) show a higher risk of disease.
4	Disease-positive patients tend to have higher BP and BMI values together.
5	The dataset is clean with no missing values or duplicate records.

Conclusion

Sugar level, BMI, and blood pressure together are strong indicators of disease risk. Patients with higher values in these factors are more likely to have a positive disease status. Since the dataset contains no missing values or duplicate records, it is suitable for reliable analysis. Regular monitoring of sugar level, BMI, and blood pressure can help in the early detection and management of disease risk.

Question 3: Retail Sales Dataset

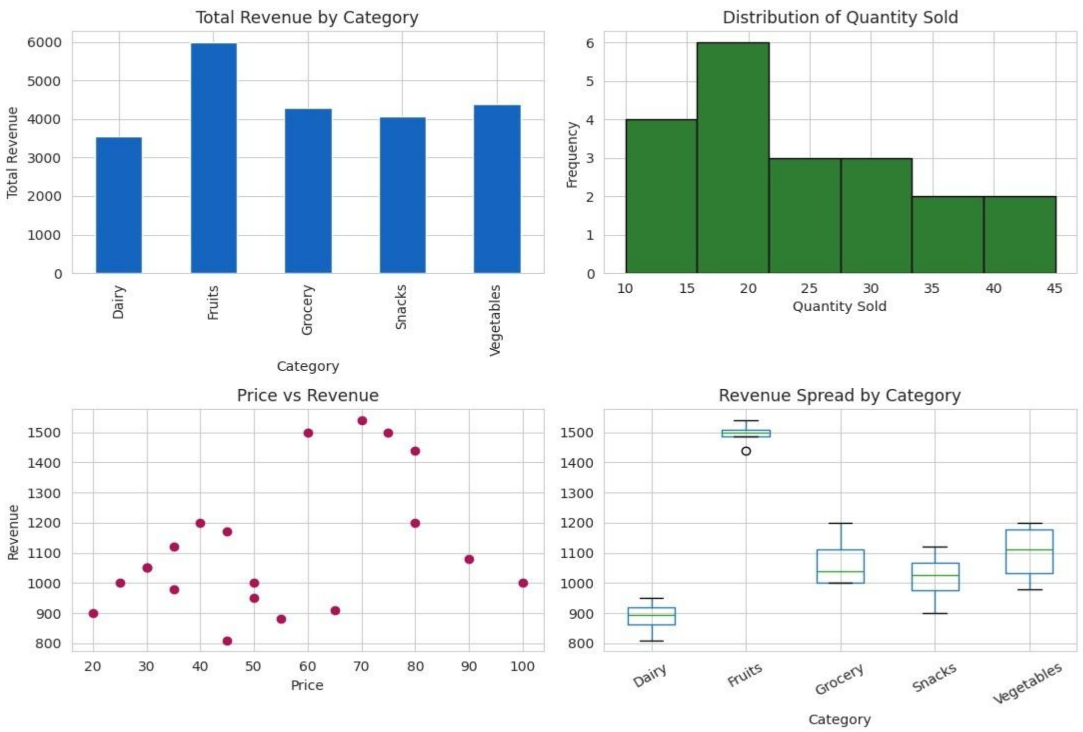
Goal: Analyze product sales based on category, price, quantity sold, discount, and revenue to identify high-selling products.

Dataset Information

Property	Value
Shape (Rows, Columns)	(20, 6)
Total Rows	20
Total Columns	6
Missing Values	0
Duplicate Rows	0

Column Data Types

Column Name	Data Type
Product_ID	str
Category	str
Price	int64
Quantity_Sold	int64
Discount	int64
Revenue	int64



Descriptive Statistics (Numeric Columns)

Statistic	Price	Quantity_Sold	Discount	Revenue
Count	20.0	20.0	20.0	20.0
Mean	54.00	24.00	4.00	1114.00
Standard Deviation (Std)	22.80	9.67	1.56	221.42

Statistic	Price	Quantity_Sold	Discount	Revenue
Minimum (Min)	20.00	10.00	2.00	810.00
25% (Q1)	35.00	17.50	3.00	972.50
50% (Median)	50.00	21.00	4.00	1050.00
75% (Q3)	71.25	30.50	5.00	1200.00
Maximum (Max)	100.00	45.00	7.00	1540.00

Observations

Observation No.	Description
1	The Fruits category generates the highest revenue overall.
2	Snacks have the highest quantity sold but a lower price per unit.
3	A higher price does not always result in higher revenue because quantity sold also has a major impact.
4	Grocery items have higher prices but lower quantities sold.
5	No missing values or duplicate records were found in the dataset.

Conclusion

Revenue depends on both price and quantity sold rather than on price alone. Products with a balanced combination of selling price and sales volume generate higher revenue. Fruits and Vegetables are the strongest revenue-generating categories. Since the dataset contains no missing values or duplicate records, it is suitable for reliable retail sales analysis.

Question 4: Banking Loan Approval Dataset

Goal: Analyze loan approval decisions based on income, credit score, loan amount, and employment type.

Dataset Information

Property	Value
Shape (Rows, Columns)	(20, 6)
Total Rows	20
Total Columns	6
Missing Values	0
Duplicate Rows	0

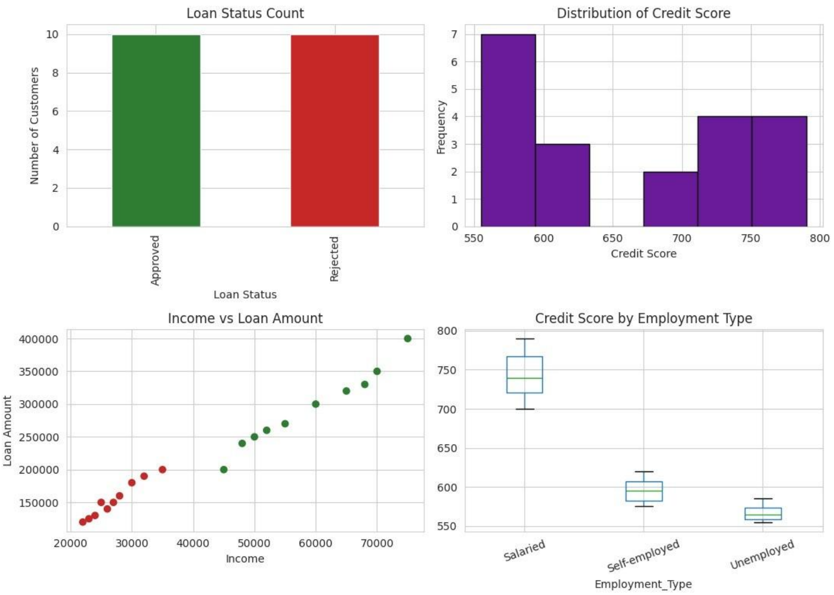
Column Data Types

Column Name	Data Type
Customer_ID	str
Income	int64
Credit_Score	int64
Loan_Amount	int64
Employment_Type	str
Loan_Status	str

Descriptive Statistics (Numeric Columns)

Statistic	Income	Credit_Score	Loan_Amount
Count	20.0	20.0	20.0
Mean	43000.00	664.00	223250.00
Standard Deviation (Std)	17923.82	85.53	83733.21
Minimum (Min)	22000.00	555.00	120000.00
25% (Q1)	26750.00	583.75	150000.00
50% (Median)	40000.00	660.00	200000.00
75% (Q3)	56250.00	735.00	277500.00

Statistic	Income	Credit_Score	Loan_Amount
Maximum (Max)	75000.00	790.00	400000.00



Observations

Observation No.	Description
1	Customers with credit scores above 700 are mostly approved for loans.
2	Salaried employees have a much higher loan approval rate than self-employed or unemployed applicants.
3	Customers with higher incomes tend to receive approval for larger loan amounts.
4	Unemployed customers are rejected in almost all cases in this dataset.
5	No missing values or duplicate records were found in the dataset.

Conclusion

Credit score and employment type are the strongest factors influencing loan approval, followed by income level. Applicants with high credit scores, stable salaried employment, and higher incomes have a greater chance of loan approval and are more likely to receive larger loan amounts. The dataset contains no missing values or duplicate records, making it suitable for reliable analysis.

Question 5: Agriculture Crop Yield Dataset

Goal: Analyze crop yield based on rainfall, temperature, fertilizer usage, and soil type.

Dataset Information

Property	**Value**
Shape (Rows, Columns)	(20, 6)
Total Rows	20
Total Columns	6
Missing Values	0
Duplicate Rows	0

Column Data Types

Column Name	**Data Type**
Farm_ID	str
Rainfall_mm	int64
Temperature	int64
Fertilizer_kg	int64
Soil_Type	str
Crop_Yield_kg	int64

Descriptive Statistics (Numeric Columns)

Statistic	**Rainfall_mm**	**Temperature**	**Fertilizer_kg**	**Crop_Yield_kg**
Count	20.0	20.0	20.0	20.0

Statistic	Rainfall_mm	Temperature	Fertilizer_kg	Crop_Yield_kg
Mean	743.00	29.55	54.75	2820.00
Standard Deviation (Std)	144.34	2.76	10.51	681.41
Minimum (Min)	500.00	25.00	40.00	1800.00
25% (Q1)	615.00	27.75	45.75	2175.00
50% (Median)	770.00	29.00	54.50	2975.00
75% (Q3)	857.50	32.00	60.50	3250.00
Maximum (Max)	960.00	34.00	75.00	3900.00

Observations

Observation No.	Description
1	Higher rainfall and greater fertilizer usage are both linked to higher crop yield.
2	Sandy soil produces the lowest crop yield even with fertilizer application.
3	Higher temperatures are generally associated with lower crop yield due to drier conditions.
4	Loamy soil produces higher crop yields compared to other soil types.
5	No missing values or duplicate records were found in the dataset.

Conclusion

Soil type and rainfall are the major factors influencing crop yield, while fertilizer usage further improves production, especially in Loamy soil. Higher rainfall combined with appropriate fertilizer application results in better crop productivity, whereas higher temperatures tend to reduce yield. The dataset is clean, with no missing values or duplicate records, making it suitable for reliable agricultural analysis.

Overall Summary

The same Exploratory Data Analysis (EDA) process was applied across all five datasets. Each dataset was examined by checking its structure, verifying data quality, computing descriptive statistics, identifying key patterns through analysis, and drawing meaningful conclusions. This systematic EDA approach supports informed decision-making in the domains of student performance, healthcare, retail sales, banking loan approval, and agriculture.